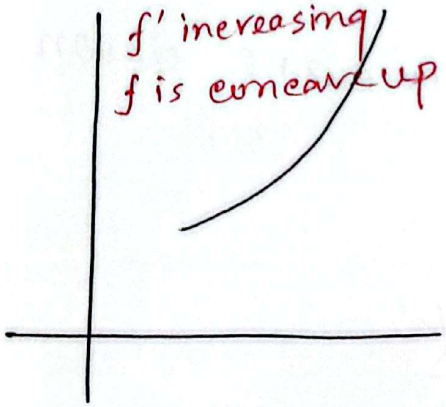
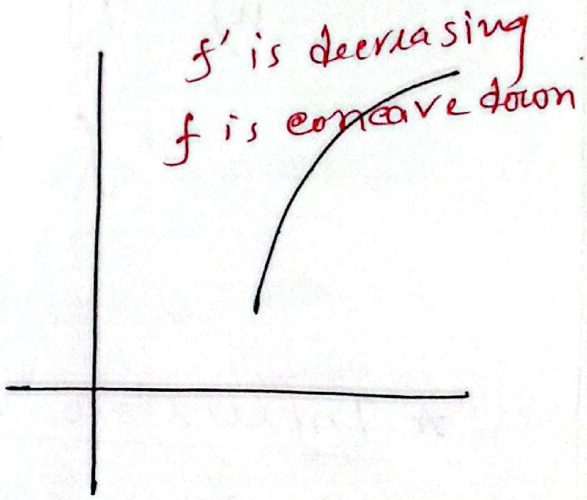
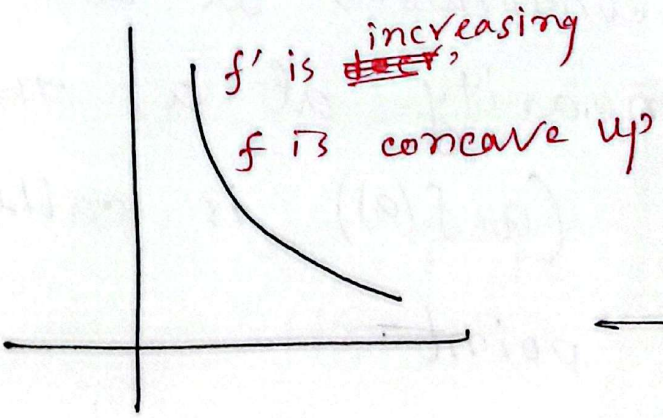
* concavity and points of Inflection:

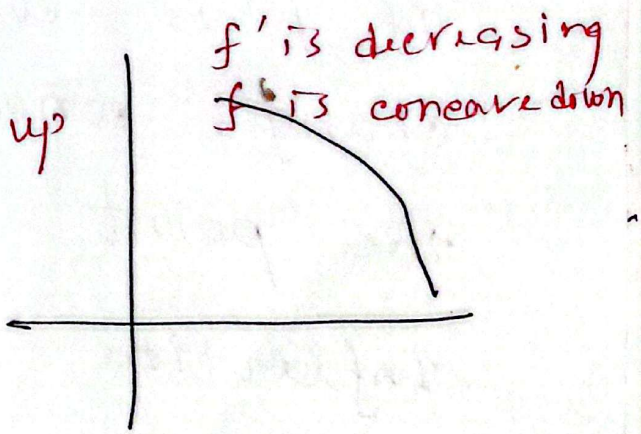
Let f be a function that is differentiable over an open Interval I. If f' is increasing over I, we say f is concave up over I. ~~If~~

If f' is decreasing over I, we say f is concave down over I.









✱ Theorem:

Let f be a function that is twice differentiable over an interval I.

i) If $f''(x) > 0$ for all $x \in I$, then f is concave up ~~at~~ over I.

ii) If $f''(x) < 0$ $\forall$ $x \in I$, then, f is concave down over I.

✱ Inflection point:

If f is continuous at a and f changes concavity at a, then the point $(a, f(a))$ is called the Inflection point.

To find Inflection point, we will calculate $f''(x) = 0$.

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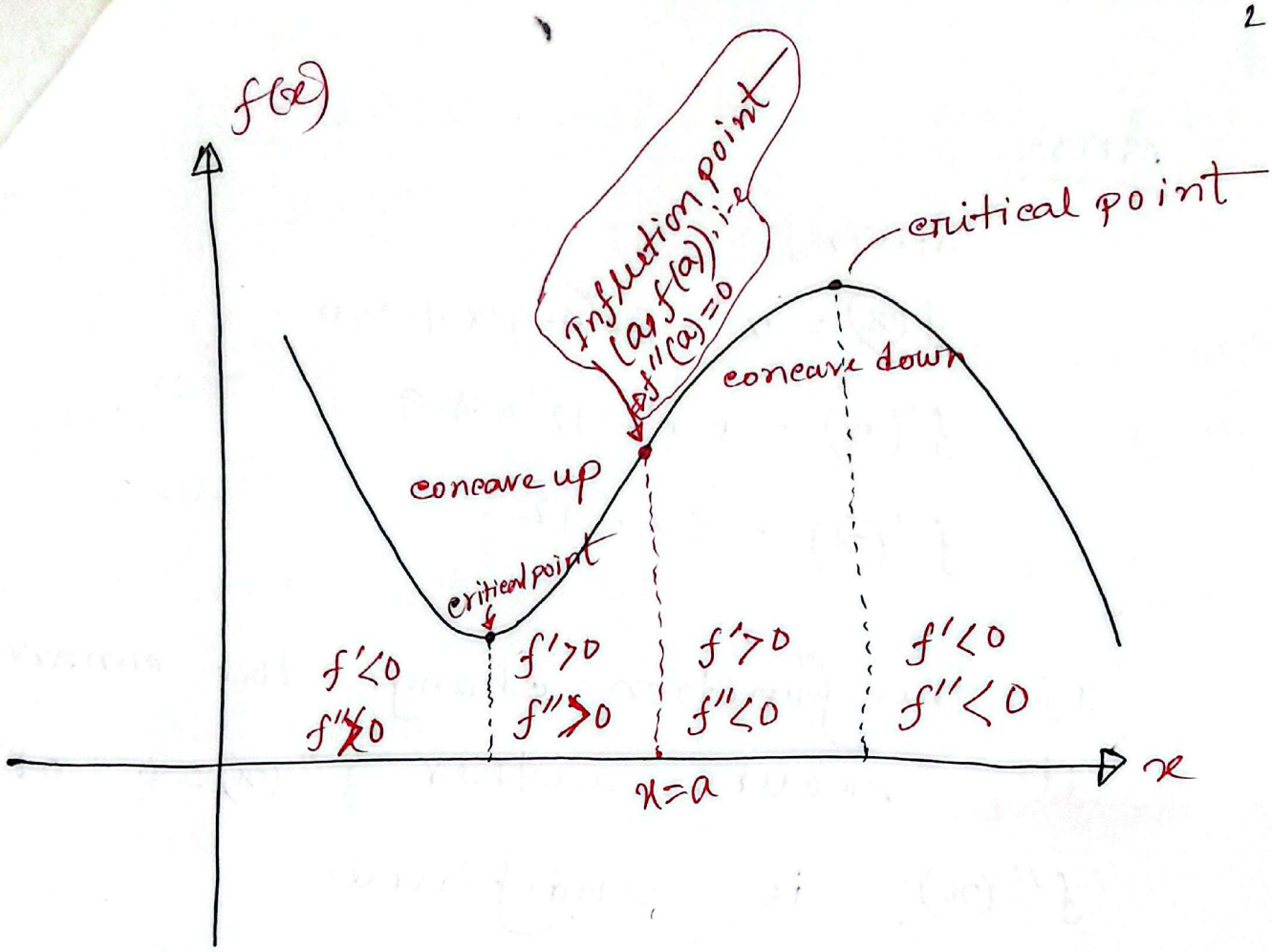


Example:

let $f(x) = x^3 - 6x^2 + 9x + 30$.

Determine all intervals where f is concave up and all intervals where f is concave down.

list all inflection points for f. Use a graph utility to confirm your results.

Ans:

Given,

$f(x) = x^3 - 6x^2 + 9x + 30$

$f'(x) = 3x^2 - 12x + 9$

$f''(x) = 6x - 12$

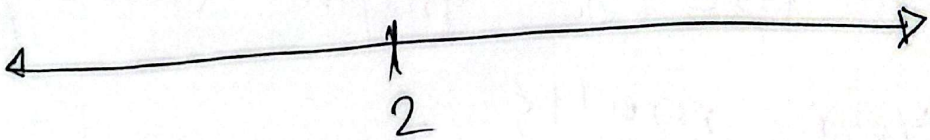
If the function change the concavity, it occurs either $f''(x) = 0$ or $f''(x)$ is undefined.

So, for Inflection points

$f''(x) = 0$

$\Rightarrow 6x - 12 = 0$

$\therefore x = 2$



Inflection points $(2, f(2)) = (2, 32)$.

$f(2) = 2^3 - 6\times 2^2 + 9\times 2 + 30 = 32$

our domain divides two intervals with respect to the inflection point at $x=2$

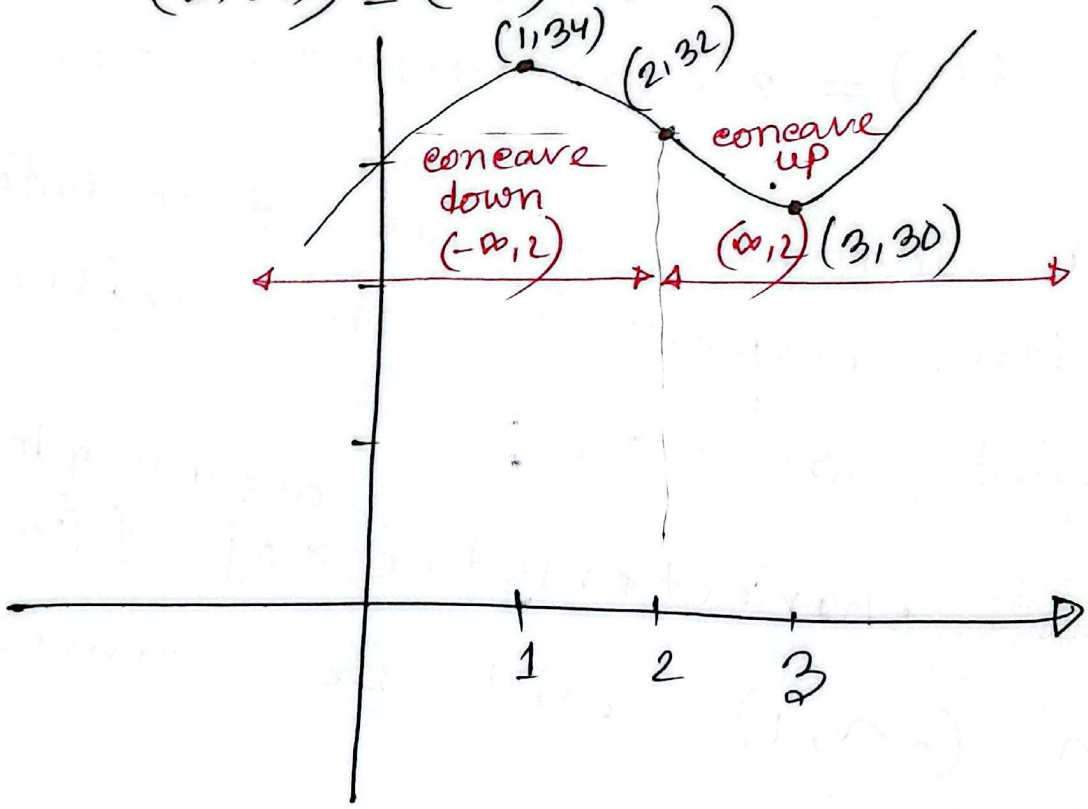
the characteristic according to concavity of $f(x)$ on $(-\infty, 2)$ will be same.

~~at~~ so, for $(2, \infty)$

Interval	Test point	sign of $f''(x) = 6x-12$	
$(-\infty, 2)$	$x=0 \in (-\infty, 2)$	$(-)$	f is concave down
$(2, \infty)$	$x=3 \in (2, \infty)$	$(+)$	f is concave up.

critical points ~~at~~ $x = 1, 3$

$(1, f(1)) \equiv (1, 34)$ & $(3, f(3)) \equiv (3,$



Taylor Series :- We define the Taylor series expansion of $f(x)$ about $x=a$ to be the series

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)(x-a)^n}{n!} + \cdots$$

$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)(x-a)^n}{n!}$$

When we truncate this series we obtain the Taylor polynomial of $f(x)$ about $x=a$ of degree n denoted by

$$P_{n,a}(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)(x-a)^n}{n!}$$

Example :- Find the Taylor series expansion of $f(x) = \cos x$ about $x = \frac{\pi}{2}$.

Solution :- Given that

$$f(x) = \cos x$$

$$f\left(\frac{\pi}{2}\right) = 0, \qquad f'''(x) = \sin x$$

$$f'(x) = -\sin x \quad f'(\pi/2) = -1 \qquad f'''(\pi/2) = 1$$

$$f''(x) = -\cos x \quad f''(\pi/2) = 0$$

Therefore,

$$\cos x = f(\pi/2) + f'(\pi/2)(x-\pi/2) + \frac{f''(\pi/2)}{2!}(x-\pi/2)^2$$

$$+ \frac{f'''(\pi/2)}{3!}(x-\pi/2)^3 + \cancel{f^{iv}} \ldots$$

$$= -(x-\pi/2) + \frac{(x-\pi/2)^3}{3!} + \ldots$$

Maclaurin series:- We define the Maclaurin series expansion of $f(x)$ about $x=0$ to be the series

$$\cancel{f(x) \simeq f(0) + f'(0)\frac{x}{2!} + f''(0)\frac{x^3}{3!} + f'''(0)\frac{x^4}{}}$$

$$f(x) \simeq f(0) + f'(0)\frac{x}{1!} + f''(0)\frac{x^2}{2!} + f'''(0)\frac{x^3}{3!} + \cdots$$

$$\simeq \sum_{n=0}^{\infty} \frac{f^{(n)}(0)x^n}{n!}$$

Example:- Find the Maclaurin series expansion of $\frac{1}{1-x}$.

Solution:- Given that

$$f(x) = \frac{1}{1-x}$$

Example:- find the Maclaurin series expansion

: $f(x) = \frac{1}{1-x}$.

Solution:- Given that

$$f(x) = \frac{1}{1-x}$$

$f(0) = 1$ $\qquad f'(x) = \frac{+1}{(1-x)^2}$ $\qquad f'(0) = +1$.

$$f''(x) = \frac{2}{(1-x)^3} \qquad f''(0) = 2$$

$$f'''(x) = \frac{+6}{(1-x)^4} \qquad f'''(0) = +6 .$$

$$\frac{1}{1-x} : f(0) + f'(0)\,x + f''(0)\frac{x^2}{2!} + f'''(0)\frac{x^3}{3!} + \cdots$$

$$= 1 + x + x^2 + x^3 + \cdots$$

Ans.